

SKILL-15

Code:

```
GNU nano 7.2 skill15_redirection.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <fcntl.h>
#include <string.h>

void run_test(int order)
{
    int pipefd[2];

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    pid_t pid = fork();

    if (pid == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid == 0)
    {
        Read 182 lines
    }
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ ls -l skill15_redirection.c skill15_redirection
-rwxr-xr-x 1 bhardwaj bhardwaj 16520 Oct  7 17:16 skill15_redirection
-rw-r--r-- 1 bhardwaj bhardwaj 2012 Oct  7 17:16 skill15_redirection.c
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ ./skill15_redirection
=====
SKILL-15: REDIRECTION AND PIPE TEST
=====

TEST 1: stdout and stderr merged using 2>&1
-----
Child: standard output message
Child: standard error message
Parent: received 61 bytes through pipe

TEST 2: stdout redirected, stderr kept separate
-----
Child: standard error message
Child: standard output message
Parent: received 31 bytes through pipe

Skill-15 execution completed successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ |
```

Test actual shell redirection:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ echo "ERROR MESSAGE" >&2
ERROR MESSAGE
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ echo "OUTPUT MESSAGE" > output.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ cat output.txt
OUTPUT MESSAGE
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ |
```

Test merging stdout and stderr:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ ls /valid_path /invalid_path > merged.txt 2>&1
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ cat merged.txt
ls: cannot access '/valid_path': No such file or directory
ls: cannot access '/invalid_path': No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ |
```

Test ordering of redirection:

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ ls -l 2>&1 | wc -l
6
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ ls /does_not_exist > order1.txt 2>&1
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ cat order1.txt
ls: cannot access '/does_not_exist': No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ ls /does_not_exist 2>&1 > order2.txt
ls: cannot access '/does_not_exist': No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ cat order2.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ |

```

Check the generated files:

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ ls -l output.txt merged.txt order1.txt order2.txt
-rw-r--r-- 1 bhardwaj bhardwaj 120 Oct  7 17:19 merged.txt
-rw-r--r-- 1 bhardwaj bhardwaj  63 Oct  7 17:20 order1.txt
-rw-r--r-- 1 bhardwaj bhardwaj   0 Oct  7 17:20 order2.txt
-rw-r--r-- 1 bhardwaj bhardwaj  15 Oct  7 17:18 output.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ cat order1.txt
ls: cannot access '/does_not_exist': No such file or directory
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ cat order2.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-15$ |

```